

ABSCISSA Was The Clue For K3

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I believe the kryptos alphabet and table on the sculpture was the clue for K1, and the key for K1 was the clue for K2, and the key for K2 was the clue for K3. I will explain the clue for K3.

A cryptanalyst looking at the bottom section of Kryptos for the first time would find that the index of coincidence (IoC) of the first 336 characters matches the English language. They would assume a transposition cipher was used, and start looking for clues for the method or route. K2 provided the clue ABSCISSA.

ab·scis·sa

1. the horizontal or x-coordinate of a point in a two-dimensional system of Cartesian coordinates

The clue is straight forward, abscissa is an x-coordinate or horizontal value. The clue is similar to a **commencement word** or **blind word** like ones used in a Union Route Cipher or Stager Cipher. They often signify the dimensions of a matrix and or a route to follow. They communicate a route or value in different ways, either by a predefined list of words and routes, or by the number of characters the word has. In this case abscissa has 8 characters. **This indicates the value of X is 8.** This value seems to be the horizontal step of a skip cipher, which would also be like the diameter of the rod in a scytale cipher.

Since the ciphertext is an even 336 characters it should completely fill a matrix, without empty cells, as long as the product of the width and height is 336. The ciphertext for K3 is in a 31x11 matrix on the sculpture which equals 341 cells. That is 5 cells too large. The next size down that fits the ciphertext, where the product of the width and height is 336, and has either a width or height divisible by 8, is 24x14. See below.

Now all the cells with an index divisible by 8 on the X axis are aligned into columns and highlighted in gray. This helps count by 8. But as you see above, reading every 8th character left to right outputs OOIAUL which is meaningless. At this point we would look to see if the matrix was rotated by checking 8ths on the Y axis too. On the right side, counting every 8th character downwards reads SLOWLY. There we have the solution! It is clear that the matrix is rotated 90 degrees clockwise. We can confirm by rotating counterclockwise (as shown on page 1) and placing it into a matrix with a width divisible by 8 (as shown below).

This proves that K3 is a once rotated skip cipher of value 8 and ABSCISSA was the key.